

Class 16 Dihedral Groups

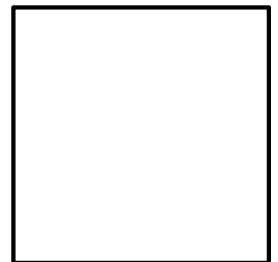
Goals

1. Understand structure of dihedral groups.
2. Be able to determine the order of a group and the order of an element in a group.
3. Be able to determine the center of a group and the centralizer of an element.

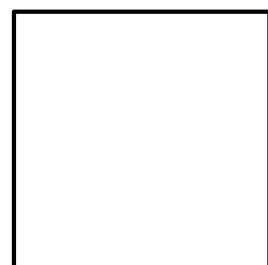
Definition: The dihedral group, D_n , is

Example:

Counterclockwise rotations



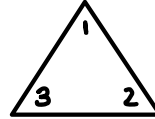
Reflections



Dihedral Group Class Activity

Students break up into pairs or triples.

Each group is given an equilateral triangle. Label corners 1, 2, 3 in clockwise order with 1 at the top, 2 on the right and 3 on the left



List all symmetries of the triangle.

Symmetries of Triangle – Dihedral Group D_3	Permutations of $\{1,2,3\}$ – Symmetric Group S_3

Cayley Table for D_3

	R_0	R_{120}	R_{240}	F_1	F_2	F_3
R_0						
R_{120}						
R_{240}						
F_1						
F_2						
F_3						

Cayley Table for S_3



The order of a group, G , is

Examples

$$|D_3| =$$

$$|D_4| =$$

$$|D_n| =$$

$$|\mathbf{Z}_n| =$$

$$|U(10)| =$$

$$|S_n| =$$

The order of an element, g , is

Examples

$$\text{In } S_5, |(12) (345)| =$$

$$\text{In } D_4, |R_{270}| =$$

$$\text{In } \mathbf{Z}_{10}, |6| =$$

$$\text{In } U(10), |9| =$$

